Robust Sliding Mode Control of a Rotary Hook

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Abstract: Nowadays, boom cranes are highly automated machines with which numerous researches have been devoted to designing and implementing automatic controllers to suppress the residual vibrations during the payload transloading process. As an essential part of the boom crane system, rotary hook device plays an important role in adjusting the skew angle of the payload to match the skew angle of vessels or container trucks. However, at present, the rotary hook is still manually controlled by highly experienced and skilled crane operators because of its under-actuated characteristic. Therefore, in this paper, a robust Sliding Mode Controller (SMC) will be developed for an uncertain rotary hook system which can automatically drive the payload to desired angle whilst suppressing the residual skew oscillation caused by flexible ropes. The parametric uncertainties presenting in the system parameters will be thoroughly considered in the SMC design process. By conducting the sliding mode analysis, the constraints on the controller gains are formulated to ensure the robust stability of the closed-loop system. Moreover, a conventional Linear Quadratic Integral (LQI) controller will be introduced as a benchmark to compare with the robust SMC. All of conclusions will be drawn from both simulation and experimental results.

Keywords: sliding mode control, coupled sliding surface, LQI, parametric uncertainty, indirect Lyapunov method.

1. INTRODUCTION

Automation has become popular as a trend for modern harbours, and as a consequence, vibration suppression for crane systems have been placed in the forefront because of its urgency. Usually, the payload is suspended below the boom tip (in boom crane) or trolley (in overhead crane) by flexible ropes. This is the main cause of residual oscillations when the payload is transferred from one point to another in the workspace or when the skew orientation is changed.

In order to highlight the necessity of payload skew angle adjustment, consider the shore to ship loading process in Fig. 1a, where the payload is a container which can lie on the ground at an arbitrary skew angle γ as such, the loading process can only be proceeded if the skew angle of the container matches the skew angle of the vessel. A similar issue is also encountered in ship to shore unloading processes (as shown in Fig. 1b) where the container must be rotated $180° - \gamma$ in order to make the container door locate at the rear end of the truck. In reality, a device called rotary hook is utilized to rotate the payload to any desired skew angle. In fact, as described in Fig. 2, the spreader grips the container by twist-lock mechanism and also be connected with a motor shaft which is mounted inside the housing of the rotary hook. Finally, the rotary hook device will be hung below the crane boom tip by ropes. After the rotation of the motor shaft drives the payload to the desired angle, a large residual skew oscillation will appear because the whole system (including payload and rotary hook device) has large inertia and is suspended by flexible ropes.

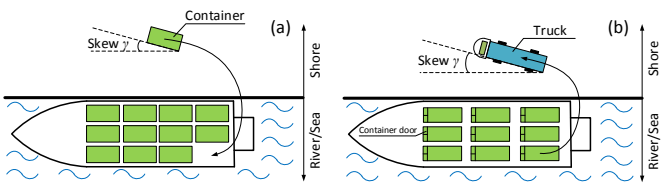

Fig. 1. Necessity of the payload skew orientation correction in the loading case (a) and the unloading case (b).

Despite the fact that many anti-swing controllers have been developed for the boom crane system (Terashima et al. (2007); Uchiyama (2009); Neupert et al. (2010)) or overhead crane system (Miyoshi and Terashima (2006, 2008); Aribowo and Terashima (2014)), there are only a few researches paying attention to designing automatic controllers for suppressing the residual skew oscillation of the payload. In fact, nowadays, the payload skew angle control through the rotary hook device is always manually undertaken by workers. Naturally, the rotary hook is an under-actuated system which has only one control input (torque generated by a motor) and two control objectives (i.e. drive the payload to desired angle and suppress the skew oscillation simultaneously) thus the crane operator

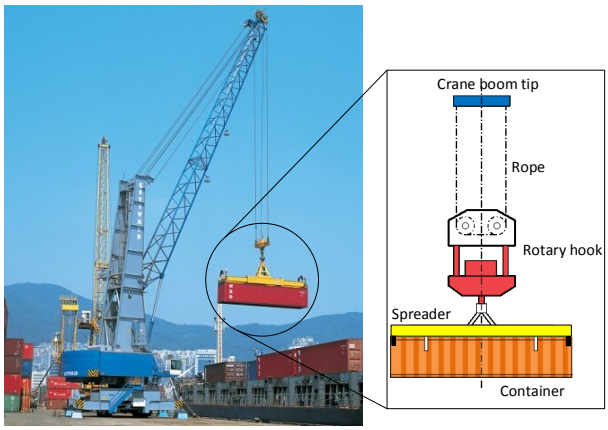

Fig. 2. Rotary hook device in the boom crane system.

needs significant experience and, as a result, the transfer process is not particularly time efficient. Therefore, in order to reduce the workload for the crane operator as well as decrease the duty cycle time, an automatic controller should be developed to meet the need of the harbor industry.

The very first study on the rotary hook system can be traced to Gustafsson (1993), where the mathematical model of the rotary hook system is derived when the payload is considered as a container. Two types of controllers have been developed: time-optimal and state feedback control based on a simplified double integrator plant. Both controller types are designed with the assumption that the natural frequency term of the system is perfectly known, which is unrealistic in practice. Full-scaled experiments are realized to verify the performance of the state feedback controller where the controller gains are chosen based on trial and error. Recently, Schaper et al. (2014) proposed a 2-DOF skew control with novel trajectory generation and a state feedback control for the rotary hook system. The feed-forward path is online computed in real time based on the control input obtained from simulation. Similar to the work of Gustafsson (1993), a conventional state feedback control is applied, however this control law is now shown to be able to stabilize the nonlinear plant by using the scalar circle criterion. The proposed controller performance is validated with test drives on a full-size mobile harbor boom crane. From our observation, none of the existing researches consider the uncertainty of the rotary hook system to an elaborate degree. In the actual working condition, the payload mass, payload inertia and rope length, widely vary between $10 - 50$ tons, $38103 - 819512$ kgm^2, and $10 - 30$ m respectively. Without the robustness consideration, such large parametric uncertainties will seriously affect the performance of controller or even cause instability to the closed-loop system. Therefore this paper aims to develop a well-known robust sliding mode controller to handle the heavy parametric uncertainties in the rotary hook system.

Sliding mode control is well-recognized due to its excellent robustness property (Fridman et al. (2011); Chin and Wheeler (2013); Al-Ghanimi et al. (2015); Rezkallah et al. (2017); Liu et al. (2017)). Fallaha et al. (2011) introduced a new exponential reaching law for both SISO and MIMO systems, which allow SMC to be robust with bounded uncertainties. Islam and Liu (2011) suggested a novel sliding mode control for a robot manipulator which can guarantee the tracking performance under large uncertainties. Among a finite set of candidates, the parameter estimation of a conventional adaptive SMC will be reset into a sub-model which most accurately describes the system. However, aforementioned SMCs are only applicable for full-actuated system. Bartolini et al. (2002) proposed a novel second order sliding mode control for a 2D container crane which guarantees the zero-swing at the end of transportation as well as active swing-damping during the load movement. Despite the theoretical section is rigorously confirmed, the experiment study are only conducted in a few cases, especially the experimental results are not satisfactory. Recently, Ngo and Hong (2012) established a SMC for an underactuated offshore container system, which is intentionally designed to be robust with external disturbances caused by waves. According to the simulation results, the SMC outperforms the conventional PD controller. Tuan et al. (2013) employed an adaptive sliding surface in order to avoid over-estimating the switching gain. However, under the sliding mode, the swing angles are claimed to be decayed to zero simply by gravitation without any further analysis on the closed-loop system dynamic.

As aforementioned discussions therefore, in this paper, once the sliding mode is established, the closed-loop system in the presence of parametric uncertainties will be systematically analyzed based on the indirect Lyapunov method. The robust stability analysis will result in the lower bounds of SMC gains, which are independent from the uncertain system parameters. Furthermore, the experimental results of the proposed SMC on a miniature rotary hook testbed will be shown to be excellent without any gain tuning process. Therefore, the proposed SMC can be implemented on the real-size rotary hook system in use at the harbor without any difficulty. Those are also the main contributions of this paper.

The remainder of the paper are listed as follows. In section 2, a modified mathematical model of the rotary hook system is introduced and validated on the experiment apparatus. In section 3, SMC with a coupled linear sliding surface will be proposed for an uncertain underactuated rotary hook system. The random simulation results are provided in section 4, which will prove the superior of robust SMC in comparison with the conventional LQI controller. Section 5 will present the experimental results of both controllers. It will be later shown that the simulation and experiment results are in total agreement, which means that the controller gains used in the simulation can be applied directly to the real plant. Therefore, the tedious gains tuning process can be avoided completely in practice and, hopefully, this result can encourage control engineers, who usually use trial and error in their controller implementation, to change to a better approach.

2. MODELING VALIDATION

2.1 Mathematical model

This section provides a brief description on the modeling of the rotary hook system. The skew motion of the rotary

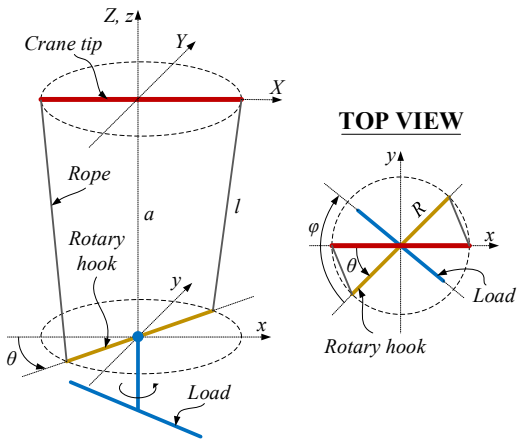

Fig. 3. Rotary hook system modeling.

Table 1. Parameters of the miniature rotary hook system

Symbol	Description
$\varphi(t)$	Relative angle between the load and the hook (rad)
$\theta(t)$	Absolute abgle of the hook versus x axis (rad)
I_L	Inertia of the payload (kgm^2)
I_H	Inertia of the hook (kgm^2)
m	Total mass of the load and the hook (kg)
R	Skewing radius (m)
l	Rope length (m)
c	Friction coefficient (Ns/rad)
g	Gravity acceleration (m/s^2)

hook system resulting from the rotation of payload is illustrated in Fig. 3. The definitions of system parameters are described in Table 1.

Using the Euler-Lagrange method and ignoring the hoisting motion, the nonlinear model of the rotary hook system is given as

$$\ddot{\theta} = f + bu, \tag{1}$$

where

$$f = \frac{1}{I_L + I_H + \dfrac{mR^4 \sin^2\theta}{a^2}} \times \left[\frac{-mR^2 \sin\theta}{a}\left(\frac{R^4\dot{\theta}^2 \sin^2\theta}{a^3} + \frac{R^2\dot{\theta}^2\cos\theta}{a} + g\right) - c\dot{\theta}\right]$$

$$b = \frac{I_L}{I_L + I_H + \dfrac{mR^4 \sin^2\theta}{a^2}}$$

$$u = \ddot{\varphi},$$

and $a = \left(l^2 - 2R^2\left(1 - \cos\theta\right)\right)^{1/2}$.

By assuming the absolute skew angle θ and square of skew rate $\dot{\theta}^2$ terms are sufficiently small, the linear model of the rotary hook system can be derived as

$$\ddot{\theta} = -\frac{mgR^2}{l\left(I_L + I_H\right)}\theta - \frac{c}{I_L + I_H}\dot{\theta} + \frac{I_L}{I_L + I_H}u. \tag{2}$$

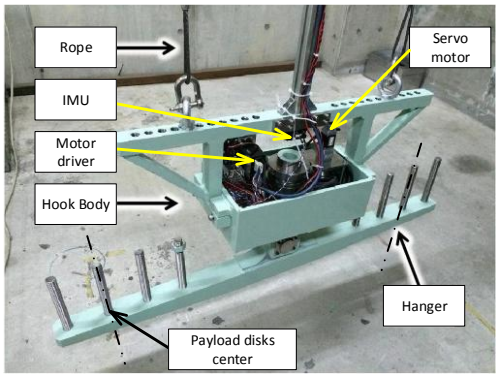

Fig. 4. Experiment testbed structure.

Note that in both nonlinear and linear models, the control input is an acceleration $\ddot{\varphi}$ which is generated by a servo motor, hence the system is clearly underactuated since there are two control objectives should be achieved (i.e. $\varphi \to \varphi_d$, $\theta \to 0$ with φ_d is the reference of φ).

2.2 Experiment apparatus

Our ultimate goal is to implement the designed controllers on real rotary hook system at the harbor. In the first step to achieve the final goal, an experiment testbed is built in the laboratory. The structure of experiment apparatus is illustrated in Fig. 4. The actuator is a TAMAGAWA TS4738 AC servo motor. Based on the maximum torque that the motor can provide and the maximum inertia of payload, the control input is approximately limited at $|\ddot{\varphi}_{\max}| = 0.4$ rad/s^2. The relative angle φ between the payload and the hook can be measured by an encoder attached to the end of motor. A 9-DOF high accuracy IMU TAMAGAWA AU7595 is utilized to measure the skew angle θ of the hook.

Note that, in the experiment testbed, the payload mass m and payload inertia I_L can be adjusted by adding/reducing the number of payload disks. Moreover, the rope can be changed to a different length manually. The variation ranges of system parameters in the experiment apparatus are given as $I_L \in [2.64, 41.89]$ (kgm^2), $I_H \in [1, 1.5]$ (kgm^2), $m \in [77.39, 207]$ (kg), $l \in [2.5, 5.5]$ (m), and $c \in [0.1, 0.5]$ (Ns/rad). Besides, the skewing radius R is fixed at 0.25 (m). The nominal of each uncertain parameter is chosen at the average of its variation range. Therefore the variation percentages of I_L, I_H, m, l, and c are $\pm 88.1\%$, $\pm 20\%$, $\pm 45.6\%$, $\pm 37.5\%$, and $\pm 66.67\%$ respectively. Note that generally, all actual system parameters cannot be measured in the real process therefore the sliding mode controller should be designed to be robust under such parametric uncertainty conditions.

For clarification, in this paper,"maximum case" refers to the case that the maximum number of payload disks are installed on the hanger thus m and I_L achieve their maximum values (all other parameters remain at their nominal values), similarly "minimum case" refers to the case that no payload disk is installed hence m and I_L achieve their minimum values and finally "nominal case" corresponds to the case that all system parameters are at their nominal values.

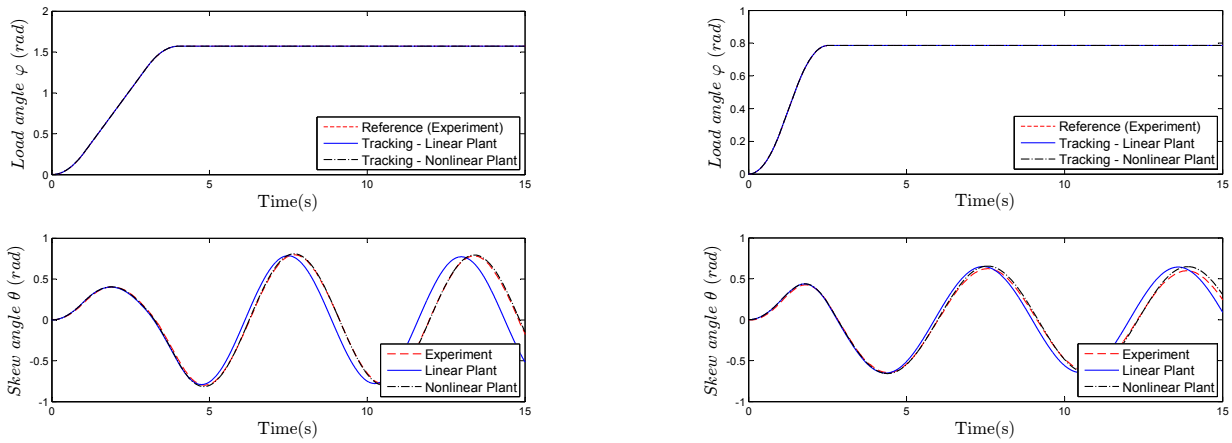

Fig. 5. Model validation in the nominal case (left) and maximum case (right).

2.3 Model validation result

The rotary hook modelling will be validated in two cases: nominal and maximum case. In these cases, the payload mass and inertia are sufficiently large hence it is easily to generate a large hook skew angle θ in order to excite the nonlinearity of the system. In each case, the actual parameters of the experiment testbed will be supplied for the nonlinear model (1) as well as the linear model (2) in the simulations.

According to the results shown in Fig. 5, the responses of the nonlinear plant perfectly match with the experiment data whereas there are significant phase shifts in the hook skew angle θ between the linear plant and the experiment results. Therefore it is concluded that the nonlinear model provides a better description on the rotary hook system dynamic. For this reason, in next section, we will propose a nonlinear SMC which employs the nonlinear model of the rotary hook system as well as conduct comparison between the robust SMC and a well-known LQI controller which utilizes the linear model (2).

3. ROBUST SLIDING MODE CONTROL DESIGN

Inspired from the work of Almutairi and Zribi (2009), we will propose the sliding mode control for an uncertain rotary hook system. Denote the tracking error term as $e = \varphi - \varphi_d$ with the assumption that φ_d is in a step form (i.e. $\dot{\varphi}_d = 0$), the linear coupled sliding surface is defined as

$$s = k_1\dot{e} + k_2 e + k_3\theta, \tag{3}$$

where k_1, k_2, and $k_3 \in \mathbb{R}$, whose lower bounds will be clarified in the robust stability analysis process. The first derivative of the sliding surface s is

$$\dot{s} = k_1(\ddot{\varphi} - \ddot{\varphi}_d) + k_2(\dot{\varphi} - \dot{\varphi}_d) + k_3\dot{\theta}. \tag{4}$$

By using the discontinuous reaching law $\dot{s} = -K\mathrm{sgn}(s)$, the control law is then derived as

$$u = k_1^{-1}(k_1\ddot{\varphi}_d - k_2\dot{\varphi} + k_2\dot{\varphi}_d - k_3\dot{\theta} - K\mathrm{sgn}(s)). \tag{5}$$

In order to show the ability of the control law (5) in pushing the system to the sliding manifold $s = 0$, a conventional Lyapunov candidate $V = (1/2)s^2$ is introduced, therefore

$$\dot{V} = \dot{s}s = \left(k_1 u - k_1\ddot{\varphi}_d + k_2\dot{\varphi} - k_2\dot{\varphi}_d + k_3\dot{\theta}\right)s. \tag{6}$$

Substituting the control law (5) into (6) yields $\dot{V} = -K|s|$, hence by choosing $K > 0$, we can guarantee that $\dot{V} \leq 0$ which implies s is bounded. The second derivative of V is

$$\ddot{V} = -K\frac{s}{|s|}\dot{s} = -K\frac{s}{|s|}\left(-K\mathrm{sgn}(s)\right) = K^2. \tag{7}$$

Therefore, $\ddot{V}$ is bounded thus according to the Barbalat lemma, it can be obtained that $\dot{V} \to 0$ as $t \to 0$ in finite time.

With an appropriate constraint on the switching gain K, the system can remain sliding on the sliding surface $s = 0$. However, because the sliding surface is a combination of the tracking error e and the residual hook skew angle θ thus it is not guaranteed that on the sliding manifold, all of the controlled variables will converge to their references (i.e. $\varphi \to \varphi_d$, $\theta \to 0$). Therefore, the following theorem will be proposed.

Theorem 1. For the uncertain rotary hook system described in (1), by using the sliding mode control law (5) with a coupled linear sliding surface (3), the sufficient condition for all controlled output variables asymptotically converge to their references (i.e. $\varphi \to \varphi_d$, $\theta \to 0$) regardless of the parametric uncertainties in all system parameters is that k_1, k_2, and k_3 are strictly positive real numbers.

Proof. When the system is attracted and remains on the sliding surface, the equation $s = 0$ holds, therefore

$$\dot{\varphi} - \dot{\varphi}_d = -k_1^{-1}k_2(\varphi - \varphi_d) - k_1^{-1}k_3\theta. \tag{8}$$

On the sliding surface, the system will reduce the order and is governed by (9)

$$\begin{cases} \ddot{\theta} = f + bk_1^{-2}k_2^2(\varphi - \varphi_d) + bk_1^{-2}k_2k_3\theta - bk_1^{-1}k_3\dot{\theta} \\ \dot{\varphi} - \dot{\varphi}_d = -k_1^{-1}k_2(\varphi - \varphi_d) - k_1^{-1}k_3\theta. \end{cases} \tag{9}$$

Define $x_1 = \theta$, $x_2 = \dot{\theta}$, $x_3 = \varphi - \varphi_d$, we obtain

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f + bk_1^{-2}k_2k_3x_1 - bk_1^{-1}k_3x_2 + bk_1^{-2}k_2^2x_3 \\ \dot{x}_3 = -k_1^{-1}k_3x_1 - k_1^{-1}k_2x_3. \end{cases} \tag{10}$$

It is easy to recognize that all controlled variables will converge to their references asymptotically if the nonlinear autonomous system defined in (10) is asymptotically stable at the equilibrium point (0,0,0).

Based on the indirect Lyapunov method, the proof can be further proceeded. As the first step, define the state vector as $\boldsymbol{x} = [x_1 \ x_2 \ x_3]^\top$ and linearized model of the system (11) can be described as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}, \tag{11}$$

where $\boldsymbol{A} = \left.\dfrac{\partial \boldsymbol{F}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=0}$ and

$$\boldsymbol{F} = \begin{bmatrix} x_2 \\ f + bk_1^{-2}k_2k_3x_1 - bk_1^{-1}k_3x_2 + bk_1^{-2}k_2^2x_3 \\ -k_1^{-1}k_3x_1 - k_1^{-1}k_2x_3 \end{bmatrix}.$$

After linearization process for f and b terms, the system matrix $\boldsymbol{A}$ can be obtained as follows

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\alpha_1 & -\alpha_2 & -\alpha_3 \\ -k_3/k_1 & 0 & -k_2/k_1 \end{bmatrix}, \tag{12}$$

where
$\alpha_1 = \omega_n^2 - \bar{b}k_1^{-2}k_2k_3$
$\alpha_2 = c_N + \bar{b}k_1^{-1}k_3$
$\alpha_3 = \bar{b}k_1^{-2}k_2^2,$
and $\omega_n^2 = \dfrac{mgR^2}{l(I_L + I_H)}$, $c_N = \dfrac{c}{I_L + I_H}$, $\bar{b} = \dfrac{I_L}{I_L + I_H}$.

The characteristic equation associated with the system matrix $\boldsymbol{A}$ is $\det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0$, which yields

$$\lambda^3 + \left(\frac{k_2}{k_1} + \alpha_2\right)\lambda^2 + \left(\alpha_1 + \frac{k_2}{k_1}\alpha_2\right)\lambda + \frac{k_2}{k_1}\alpha_1 + \frac{k_3}{k_1}\alpha_3 = 0. \tag{13}$$

The Hurwitz matrix corresponding to (13) can be established as follows

$$\boldsymbol{H} = \begin{bmatrix} \alpha_2 + k_1^{-1}k_2 & k_1^{-1}k_2\alpha_1 + k_1^{-1}k_3\alpha_3 & 0 \\ 1 & \alpha_1 + k_1^{-1}k_2\alpha_2 & 0 \\ 0 & \alpha_2 + k_1^{-1}k_2 & k_1^{-1}k_2\alpha_1 + k_1^{-1}k_3\alpha_3 \end{bmatrix}. \tag{14}$$

Based on the Hurwitz stability criterion, the sufficient condition for the system defined in (11) asymptotically stable is given as

$$\begin{cases} k_1^{-1}k_2 + \alpha_2 > 0 \\ \alpha_1 + k_1^{-1}k_2\alpha_2 > 0 \\ k_1^{-1}k_2\alpha_1 + k_1^{-1}k_3\alpha_3 > 0 \\ (k_1^{-1}k_2 + \alpha_2)(\alpha_1 + k_1^{-1}k_2\alpha_2) - (k_1^{-1}k_2\alpha_1 + k_1^{-1}k_3\alpha_3) > 0. \end{cases} \tag{15}$$

It is quite straightforward to show that by choosing $k_1 > 0$, $k_2 > 0$, and $k_3 > 0$, all of inequalities in (15) are satisfied. Therefore the proof is now completed.

Remark 1. In order to reduce the chattering in the control signal, the $\mathrm{sgn}(s)$ (i.e. signum) function in the control law

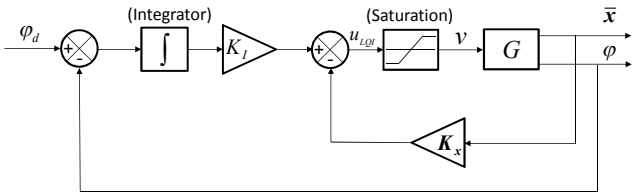

Fig. 6. LQI controller block diagram. Nominal plant of the linear rotary hook system (2) is presented by G.

(5) should be replaced by a $\mathrm{sat}(s)$ function which is defined in (16), therein ε is a constant related to the thickness of the boundary layer (Slotine and Li (1991)).

$$\mathrm{sat}(s) = \begin{cases} 1 & \text{if} \quad |s/\varepsilon| > 1 \\ s/\varepsilon & \text{if} \quad |s/\varepsilon| < 1. \end{cases} \tag{16}$$

In order to compare with the robust SMC, a conventional LQ controller with integral action will also be applied to the rotary hook system. The design procedure is adopted from Araar and Aouf (2014). By utilizing the linear model of the rotary hook in (2), the LQI control law is given as

$$u_{LQI} = K_I \int_0^t (\varphi - \varphi_d)dt - \boldsymbol{K_x}\bar{\boldsymbol{x}}, \tag{17}$$

where $\bar{\boldsymbol{x}} = \left[\varphi \ \dot{\varphi} \ \theta \ \dot{\theta}\right]^\top$. The block diagram of LQI controller is described in Fig. 6. Note that the controller gains K_I and $\boldsymbol{K_x}$ will be designed at the nominal system parameters and are given in next section.

4. SIMULATION RESULTS

In all simulation and experiment studies, the controller gains of SMC are fixed at $K = 0.55$, $k_1 = 4.5$, $k_2 = 1.8$, $k_3 = 4$, and $\varepsilon = 0.05$. The feedback gain and integral gain of LQI controller are designed as $\boldsymbol{K_x} = [8.00 \ 18.25 \ 14.84 \ -7.84]^\top$ and $K_I = 1.53$ respectively.

In reality, the payload mass, payload inertia, and rope length are unknown therefore random simulations are reasonably to be conducted. Both SMC and LQI control performances will be evaluated in 30 random sets of system parameters. Fig. 7 shows the simulation results of both controllers where the reference load angle $\varphi_d = 90$ degrees. It is observed that the robust SMC can guarantee the robust stability as well as the performance for the controlled system under heavy parametric uncertainties, whereas the LQI controller performance is not ensured well. This can be explained by the fact that the conventional LQI controller is only designed with the nominal system parameters and its robustness is not considered in the design scheme. Through these results, it is suggested that the conventional LQ controller, as proposed by Schaper et al. (2014), is not suitable to be implemented on the real rotary hook in use at the harbor. As indicated in the simulation, the robust SMC proved to be a promising candidate which is able meet the high requirements in practice.

5. EXPERIMENTAL RESULTS

The designed SMC and LQI controllers will now be implemented on the rotary hook experiment testbed. Note that the gains of both controllers used in simulation will be

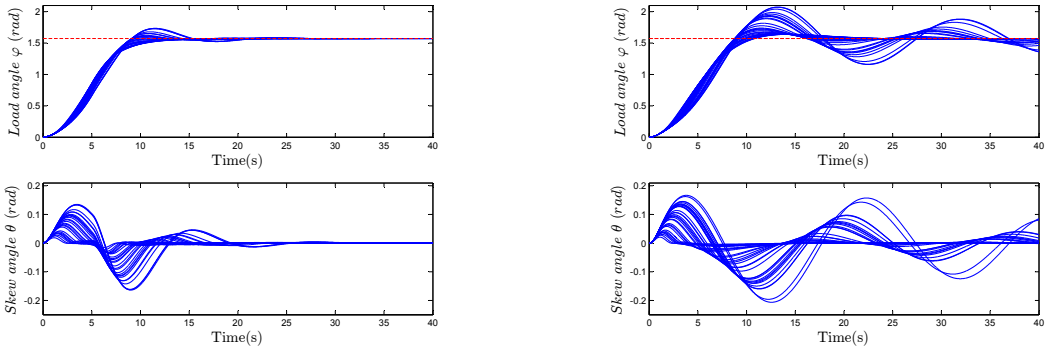

Fig. 7. Simulation result in the 30 random sets of system parameters: robust SMC (left) and LQI controller (right).

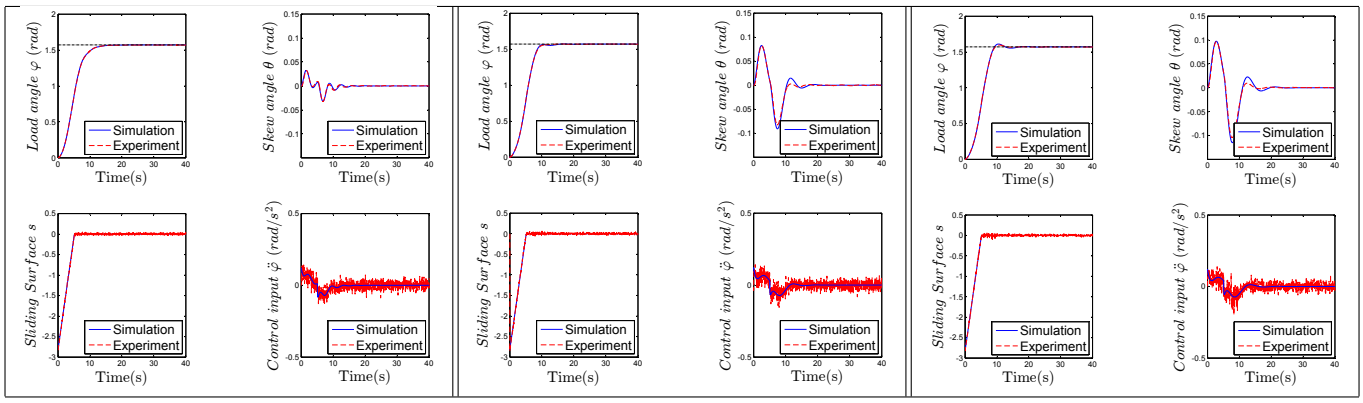

Fig. 8. Experiment result of SMC in the minimum case (left), nominal case (middle), and maximum case (right).

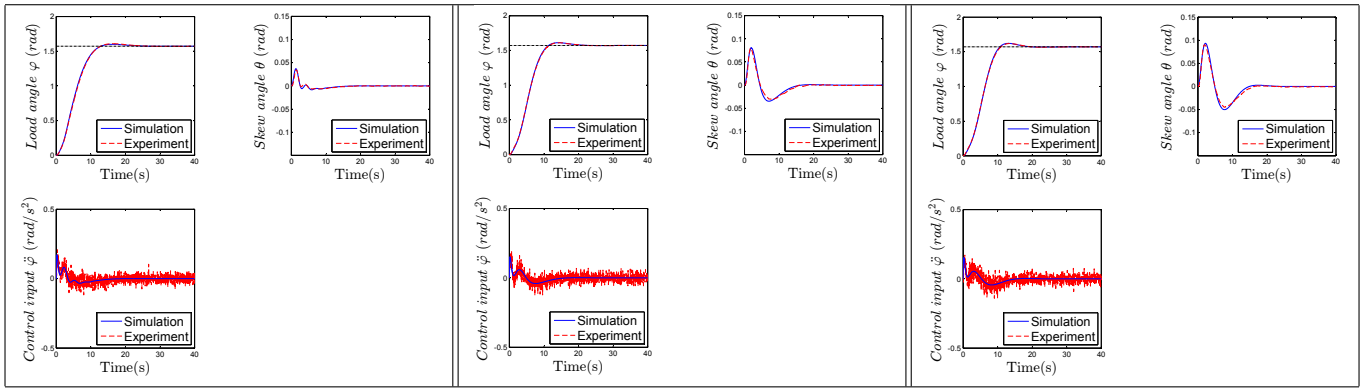

Fig. 9. Experiment result of LQI control in the minimum case (left), nominal case (middle), and maximum case (right).

applied directly to the experiment apparatus, there is not any gains tuning process in our experiments. Performances of the two controllers will be evaluated in three cases: minimum, nominal and maximum case. The experiment results are shown in Fig. 8 – Fig. 9. It is recognized that in all cases, the required time to completely suppress the residual oscillation of the hook skew angle θ are the same between two controllers. However, in a same condition, the robust SMC always performs faster settling time and smaller overshoot in the responses of load angle φ. For instance, in the maximum case, robust SMC needs 8.7 seconds to drive the payload to reach 90 degrees and 18.1 seconds to maintain the payload angle φ at its steady state, whereas LQI controller requires 11.5 seconds and 20.2 seconds respectively. Moreover the overshoot of φ in the case of robust SMC is only 0.04 rad in comparison with 0.06 rad of LQI controller.

It should be emphasized that the simulation and experiment results on the experiment testbed are in complete agreement with each other, hopefully therefore, the tuning process can be avoided completely by just using the gains in simulation when the controllers are implemented in real rotary hook system in use at the harbor.

In reality, the rotary hook system suffers from many types of disturbances surrounding its workplace environment such as from wind and especially from the collision with other containers. Therefore a non-zero initial condition of hook skew angle θ can exist and as described by the experiment result in Fig. 10, this issue will lead to significant performance degradation of the LQI controller, however robust SMC still guarantees the control performance under such condition. As our investigation discovered, this problem is generated by the wind-up phenomenon. The closed-

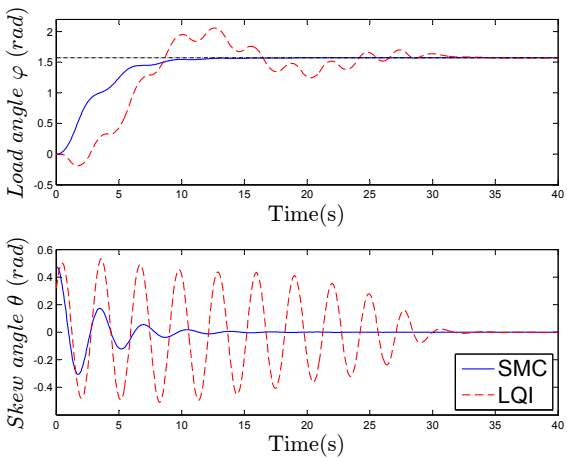

Fig. 10. Experiment result with non-zero initial condition in the minimum case.

loop system, when the LQI controller is employed, has both integral and saturation actions (as shown in Fig. 6) thus the wind-up phenomenon easily occurs when there is a large non-zero condition or disturbance. We have already eliminated this wind-up issue in the LQI controller by using the back-calculation compensation method, however the details are not mentioned here because it is out of the scope of this paper.

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